

Experiment 2: Bayes Decision Theory

Aim

To classify SMS messages as Spam or Ham using the Naive Bayes algorithm.

Tools Required

- Python
- Pandas
- VS Code

Code

```
import pandas as pd

from sklearn.feature_extraction.text import CountVectorizer
from sklearn.naive_bayes import MultinomialNB

data = {
    "label": [
        "ham", "ham", "ham", "ham", "ham",
        "spam", "spam", "spam", "spam", "spam"
    ],
    "message": [
        "How are you?",
        "Let's meet tomorrow.",
        "Are you coming to class?",
        "Call me when you reach home.",
        "Can we have lunch today?",
        "Congratulations! You won a free lottery.",
        "Claim your cash prize now.",
        "Win a free mobile phone.",
        "Exclusive offer! Click here.",
        "You have won a $1000 gift card."
    ]
}

df = pd.DataFrame(data)

vectorizer = CountVectorizer()

X = vectorizer.fit_transform(df["message"])
```

```
y = df["label"]

model = MultinomialNB()

model.fit(X, y)

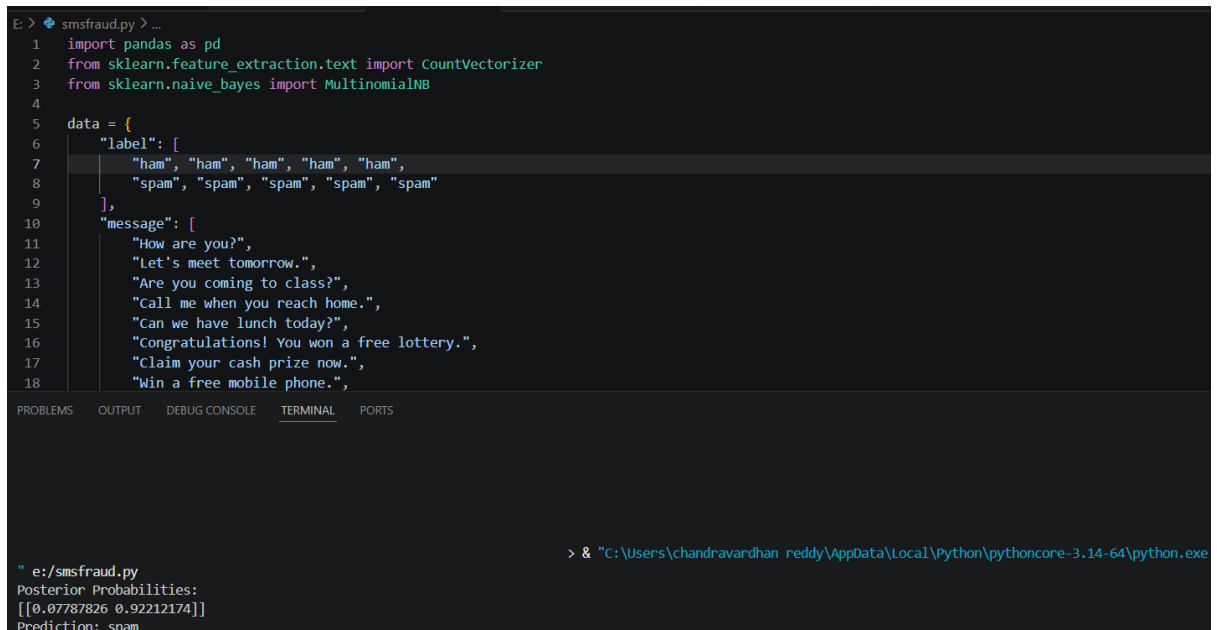
test_message = ["Congratulations! You won a free lottery ticket"]

test = vectorizer.transform(test_message)

print("Posterior Probabilities:", model.predict_proba(test))

print("Prediction:", model.predict(test)[0])
```

Output



```
E: > smsfraud.py > ...
1 import pandas as pd
2 from sklearn.feature_extraction.text import CountVectorizer
3 from sklearn.naive_bayes import MultinomialNB
4
5 data = {
6     "label": [
7         "ham", "ham", "ham", "ham", "ham",
8         "spam", "spam", "spam", "spam", "spam"
9     ],
10    "message": [
11        "How are you?",
12        "Let's meet tomorrow.",
13        "Are you coming to class?",
14        "Call me when you reach home.",
15        "Can we have lunch today?",
16        "Congratulations! You won a free lottery.",
17        "Claim your cash prize now.",
18        "Win a free mobile phone."
19    ]
20 }
21
22 vectorizer = CountVectorizer()
23 X = vectorizer.fit_transform(data["message"])
24 y = data["label"]
25
26 model = MultinomialNB()
27 model.fit(X, y)
28
29 test_message = ["Congratulations! You won a free lottery ticket"]
30 test = vectorizer.transform(test_message)
31
32 print("Posterior Probabilities:", model.predict_proba(test))
33 print("Prediction:", model.predict(test)[0])
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
> & "C:\Users\chandravardhan_reddy\AppData\Local\Python\pythoncore-3.14-64\python.exe
" e:/smsfraud.py
Posterior Probabilities:
[[0.07787826 0.92212174]]
Prediction: spam
```

Result

Bayes Decision Theory was successfully implemented using the Naive Bayes classifier, and the SMS message was correctly classified as Spam.